SAQUIB HUSSAIN

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SKILLS

- **Programming Languages**: Python, R, C, C++, DSA (C++), MySQL, HTML
- **Libraries and Frameworks**: Scikit-learn, NumPy, Pandas, Matplotlib, Seaborn, OpenCV (Image Processing), FastAPI, Streamlit.
- **Data Science**: Machine Learning, Machine Learning Algorithms, Reinforcement Learning, NLP, Deep Learning (ANN, CNN, RNN, Seq2Seq Models), Machine Learning Statistics, Data Science pipeline (data cleansing, visualization, modeling, interpretation).
- **Other Skills:** Hypothesis Testing, OOP, Excel, Git, GitHub.

WORK EXPERIENCE

Associate Engineer – R&D (AI) – Senrysa Technologies Limited (Kolkata) Sep 2025 - Present

- Engineered a secure facial recognition attendance system integrating 2 core technologies (SVM and MediaPipe) with eye-blink liveness detection and a centralized administrative dashboard.
- Architected a real-time web scraping pipeline spanning 22 state eProcurement portals, deploying a custom-trained CRNN model to automate CAPTCHA resolution for continuous metadata and document extraction.
- Developed a multi-model RAG application utilizing FAISS and Hugging Face LLMs to execute semantic searches across 3 distinct data modalities (text, tables, images) within PDFs, featuring dynamic parameter tuning and multi-format downloads via a streamlit UI.

Research Intern (Reinforcement Learning and Deep Learning) – IIT Jammu Feb 2024 – Jul 2024

- Researched and developed a Reinforcement Learning-Based Intrusion Response System (RL-IRS). Implemented the VarDetect framework, generated attacker/non-attacker datasets (Fashion MNIST, CIFAR-10), and calculate Maximum Mean Discrepancy (MMD) and Mahalanobis distances for threshold analysis.
- Deployed ResNet50 adversarial defense analysis using FGSM attacks; extracted latent embeddings via VAE and performed 3D PCA visualization. Evaluated performance achieving 97% Recall and 79% Accuracy.

Machine Learning Intern – iHUB at IIT Roorkee Jul 2023 – Sep 2023

- Implemented python libraries ML algorithms and make a project that "Explore different datasets with different classifiers"

ACHIEVEMENTS

- 2x Kaggle Expert (Datasets, Notebooks) Link
- Total 1200+ Contributions and 40+ Repositories on GitHub Link
- Completed over 800 coding challenges on various platforms Link
- Hackathon: Finalist (Awards Round) in Dark Patterns Buster Hackathon (DPBH) – IIT BHU 2023 Link

PROJECTS

Link **Object detection and tracking algorithm implementation for real-time video streams and static images**

- Designed and implemented an object detection and tracking system using YOLOv8 and Streamlit. It supported real-time video streams (RTSP, Webcam, YouTube) and static images/videos.
- Added features for task selection (detection/segmentation), adjustable confidence thresholds (25 – 100), and CSV export for detection results. Also, integrated ByteTrack and BoT-SORT algorithms for advanced object tracking and segmentation.

Link **MasterChef: Transformer-Based Recipe Generation**

- Developed a deep learning-powered application utilizing ResNet101 as an encoder and a Transformer-based decoder (8 layers, 16 head attention) to generate personalized recipes from food image.
- Supports learned (1024 positions) or sinusoidal positional encoding, with a linear and softmax final layer.

EDUCATION

Degree and Board	Institute	CGPA/Percentage	Year
B.Tech. (H) CSE (Data Science)	University Teaching Department (UTD), CSVTU, Bhilai, C.G.	7.96	2021-2025
Senior Secondary School / CBSE	Sun Valley Public School, Varanasi, U.P.	65.60%	2021
Secondary School / CBSE	J.N.V. Bhadohi, U.P.	79.80%	2019